

STRETCH TASK MANUAL

Week 1 · Advanced Entry-Level Task · Mon–Tue, Jul 6–7, 2026

Program Phase	Week 1 — Foundations
Assigned	Monday, Jul 6, 2026
Due	Tuesday, Jul 11, 2026, end of day
Tracks	Web Development AI / Machine Learning
Format	Individual submission via your team repo

This manual replaces the standard entry-level task for students who want an extra challenge. It is optional-by-request: only attempt this version if you are comfortable with the fundamentals covered in Sessions 1–2. The regular task remains the default and is graded the same way.

WEB DEV TRACK — Pixel-Perfect Responsive Build

Instead of recreating a simple webpage layout, you will rebuild a full Figma mock as a production-quality, multi-state, responsive page.

Objective

Demonstrate that you can translate a real design file into accessible, reusable, responsive code — not just a static visual copy.

Requirements

- Match the Figma mock exactly: spacing, type scale, and nested auto-layout structure — no eyeballed values.
- Implement every interactive state shown or implied in the file: hover, focus, active, disabled, loading, and error.
- Build a dark mode toggle driven by design tokens (CSS variables or a theme object), not hardcoded hex values.
- Match all breakpoints in the Figma file (mobile / tablet / desktop) with real layout reflow, not just scaling.
- Recreate any prototype transitions or micro-interactions using CSS transitions or Framer Motion.
- Build with typed, reusable components — no duplicated markup blocks.
- Run a Lighthouse accessibility audit and resolve issues until you score 100 (contrast, ARIA labels, focus order).

Deliverables

- A link to your deployed page (Vercel, Netlify, or similar) or a local run script in your README.
- Your Lighthouse accessibility report (screenshot or JSON export).
- A short README explaining your component structure and any decisions you made where the mock was ambiguous.

Time Box

One day (not two). Treat Tuesday as buffer/polish time only.

AI TRACK — Real-World Classifier Pipeline

Instead of training a classifier on a clean toy dataset, you will build an end-to-end pipeline on a messy, imbalanced, real-world dataset.

Objective

Demonstrate that you can take a dataset from raw and flawed to a justified, evaluated, explainable model — the way you would on the job.

Requirements

- Choose or receive a real-world dataset with missing values, outliers, and class imbalance (not a toy dataset).
- Perform EDA and write down your reasoning for every preprocessing decision you make.
- Handle class imbalance explicitly (e.g. SMOTE or class weighting) and justify the choice.
- Train and compare at least three model families: e.g. logistic regression, a gradient-boosted tree model, and a small neural network.
- Use proper train/validation/test splits or cross-validation — no evaluating on training data.
- Select and justify the right metric for an imbalanced problem (F1 or PR-AUC, not raw accuracy).
- Tune hyperparameters (grid search or Optuna) and log every experiment (MLflow or Weights & Biases).
- Explain your best model's predictions using SHAP or a feature-importance analysis.
- Package the work as a script or notebook that another student could re-run end-to-end and reproduce your results.

Deliverables

- Your reproducible notebook or script, plus a requirements file.
- Your experiment log (MLflow run history or W&B dashboard link/export).
- A one-page write-up: dataset, preprocessing decisions, model comparison table, final metric, and what the SHAP analysis showed.

Bonus (not required)

Wrap your trained model behind a small FastAPI endpoint so it can be queried like a real service.

Time Box

Two days, as scheduled. This task is intentionally heavier than the standard version.

How This Is Graded

- Completion of the standard task's learning goals is still the baseline — this task does not replace core competency checks.
- Partial, well-documented attempts are valued over rushed, complete ones. Show your reasoning, not just your output.
- If you get stuck, submit what you have with notes on where you stopped and why — this is treated as a genuine attempt.

Questions

Bring blockers to your track lead before end of day Monday so there's time to unblock you before the Tuesday deadline.